

“You Do It” Solution

Lecture 9: Hypothesis Testing: Two-sample Tests

Prepared as a classroom-ready detailed solution set for ECOM5005

Exercise 1: Interpreting Two-Sample Test Results

Question

The financial aid office of a university asks a sample of students about their employment and earnings. The report says that “for academic year earnings, a significant difference ($p = 0.038$) was found between gender, with men earning more on average. No difference ($p = 0.476$) was found between the earnings of black and white students.” Explain both conclusions, for the effects of gender and of race on mean earnings, in language understandable to someone who knows no statistics.

Detailed Solution

Step 1: Identify what is being compared.

Both statements concern **mean earnings in two independent groups**. For gender, the comparison is the population mean earnings of men and women. For race, the comparison is the population mean earnings of black and white students.

For each comparison, the usual two-sided null hypothesis is that the two population means are equal:

$$H_0 : \mu_1 - \mu_2 = 0, \quad H_1 : \mu_1 - \mu_2 \neq 0.$$

The p -value tells us how unusual the observed sample difference would be if there were really no population mean difference.

Step 2: Interpret the gender result, $p = 0.038$.

A p -value of 0.038 means that, **if men and women truly had the same mean academic-year earnings**, the probability of obtaining a sample difference at least as extreme as the one observed would be about

$$0.038 = 3.8\%.$$

Using the conventional significance level $\alpha = 0.05$,

$$0.038 < 0.05,$$

so we reject the null hypothesis of equal mean earnings.

The sample also shows that men’s average earnings are higher. Therefore, the data provide **statistical evidence of a gender difference in mean earnings, with men earning more on average in this sample and the estimated population difference being in the same direction**.

Step 3: Translate the gender conclusion into everyday language.

A suitable non-technical statement is:

The sample gives reasonably strong evidence that male and female students do not have the same average academic-year earnings. In this sample, men earned more on average, and the observed gap would be fairly unusual if the true average earnings of men and women were actually equal.

Step 4: Interpret the race result, $p = 0.476$.

A p -value of 0.476 is much larger than 0.05:

$$0.476 > 0.05.$$

Therefore, we **do not reject** the null hypothesis that black and white students have the same population mean earnings.

The observed black-white sample difference is quite compatible with ordinary sampling variation under equal population means. The data therefore do **not** provide convincing statistical evidence of a difference in mean earnings between black and white students.

Exercise 1: Interpretation and Final Answer

Why “No Difference” Does Not Mean “Exactly Equal”

Failing to reject H_0 for race is **not proof** that black and white students have exactly the same average earnings. It means only that this sample does not contain strong enough evidence to establish a difference. A real difference could still exist but be too small, too variable, or measured with too little data to be detected by this study.

Correct Plain-Language Interpretation

Gender: The sample provides evidence that average academic-year earnings differ between men and women. Men earned more on average in the sample, and a difference this large would occur only about 3.8% of the time from random sampling variation if the true male and female means were equal. At the 5% level, that is considered statistically significant.

Race: The sample does not provide convincing evidence that black and white students have different average academic-year earnings. The observed difference is quite plausible as ordinary random sample variation if the true population means are equal. The study therefore cannot claim a statistically detectable race difference from these data.

What the p -values Do *Not* Tell Us

- $p = 0.038$ does **not** mean there is a 3.8% probability that male and female mean earnings are equal.
- $p = 0.476$ does **not** mean there is a 47.6% probability that black and white mean earnings are equal.
- Statistical significance does not tell us whether the earnings difference is **large or practically important**. The size of the estimated difference and a confidence interval would be needed for that.
- The report describes an association between group membership and earnings. Unless the study design supports causal inference, the results should not be described as proving that gender or race **causes** earnings differences.

Answer Summary

Final answer. At the usual 5% significance level, $p = 0.038$ is small enough to conclude that the sample provides evidence of different mean earnings for men and women, with men earning more on average. By contrast, $p = 0.476$ is much too large to provide evidence of different mean earnings for black and white students. For race, the correct conclusion is **insufficient evidence of a difference**, not proof that the two population means are identical.

Final Check

For interpreting two-sample test results, remember:

- compare the p -value with the chosen significance level α ;
- if $p < \alpha$, reject the null hypothesis of equal population means;
- if $p \geq \alpha$, do not reject the null hypothesis;
- “do not reject” is not the same as “prove equal”;
- the p -value measures strength of evidence against H_0 , not the size or practical importance of the difference.